Rotary Wing Aerodynamics (AE 667)

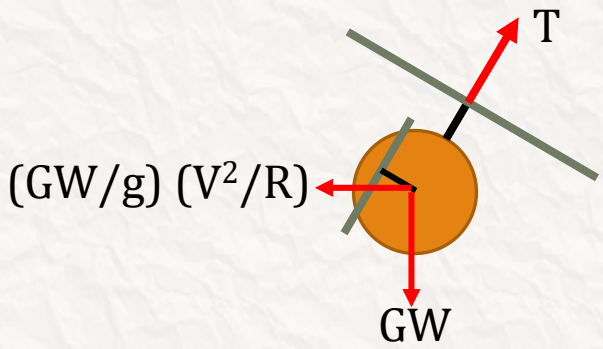


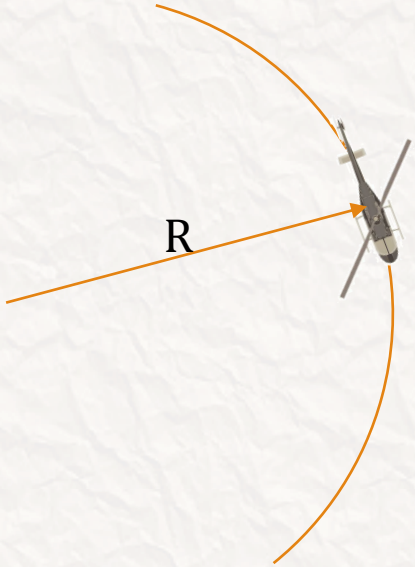
Performance at Special Flight Conditions

1. TURNS AND PULL-UPS
2. AUTOROTATION
3. MAXIMUM ACCELERATION
4. MAXIMUM DECELERATION

Load Factor in steady turn

- Load Factor (n) = Thrust / Weight
- Accounts for additional thrust needed to counter centrifugal forces for turns and maneuvers.





- $n = \sqrt{1 + \left(\frac{V\dot{\theta}}{g}\right)^2}$ $\quad \dot{\theta} = \frac{V}{R}$ $\quad R = radius\ of\ turn$
- Power calculation same as that for steady level flight. Weight multiplied by *n*

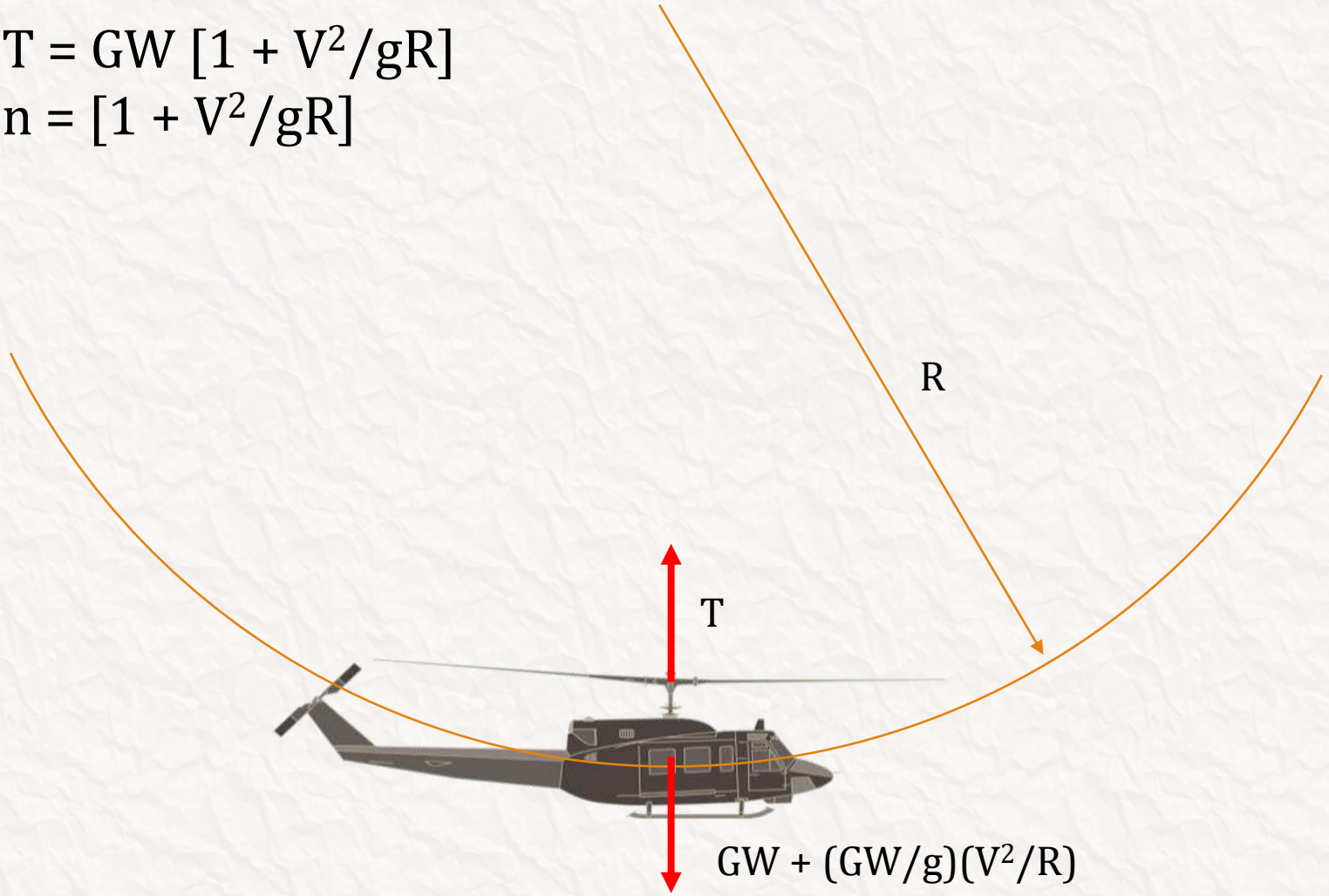
Power limited turn manoeuvre

- Engine Power augmented with exchange of Kinetic and Potential energies (Speed and Altitude) to do the turn.
- For computation:
  - Use average velocity to find the load factor n
  - Compute the kinetic and potential energy loss and subtract that from the energy required for a level turn (power × time taken in the manoeuvre)
  - The remaining energy has to be supplied by the engine. Thus, engine power can be found.

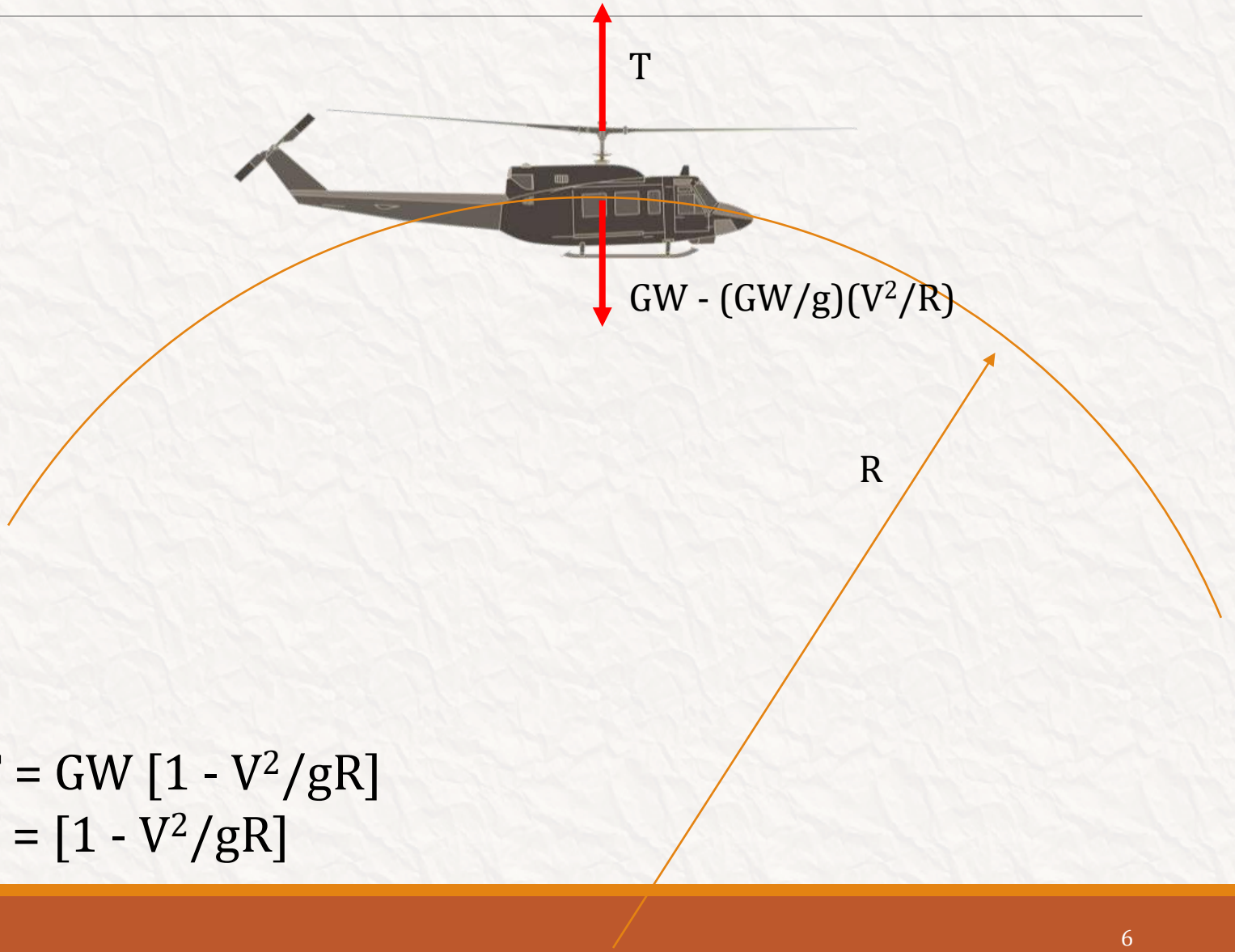
Load Factor in Pull-Up manoeuvre

$T = GW\,[1 + V^2/gR]$

$n = [1 + V^2/gR]$



Load Factor in Pull-Down manoeuvre



$T = GW\,[1 - V^2/gR]$

$n = [1 - V^2/gR]$

Autorotation

Questions to answer:

- Rate and extent of rotor speed decay (before the helicopter attains autorotative state)
- Rate of descent of the vehicle during steady autorotation
- Range of vehicle in autorotation

Inertia & time available to pilot

- Inertia of the entire rotating system = Main rotor + Tail rotor + transmission

- Total Moment of inertia =

$$J_{total} = J_{\text{Main}} + J_{Tail}\frac{\Omega_{tail}}{\Omega_{main}} + J_{transmission}$$

- Estimate of available time: $t_{KE} = \frac{K.E}{Power} = \frac{\frac{1}{2}J\Omega_0^2}{\Omega_0 Q}$

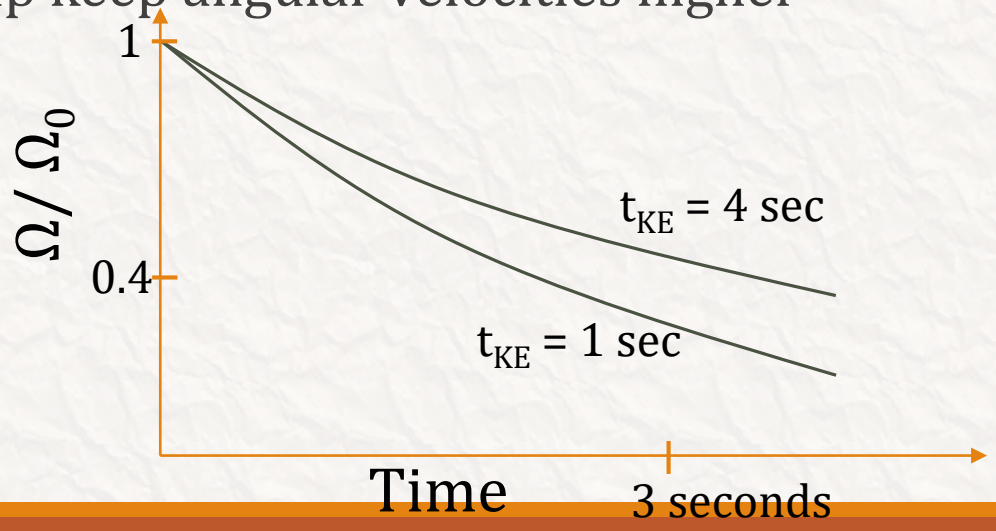
- t_{KE} is the time pilot has to reduce collective pitch to negative to enter autorotative mode. Hence, higher rotor inertia is safer.

Rotor angular velocity variation

- Rate of change of angular momentum = torque
- Assuming torque (Q) to be proportional to Ω^2,

$$J\frac{d\Omega}{dt} = -Q_0\left(\frac{\Omega}{\Omega_0}\right)^2$$

- Integrating: $\frac{\Omega}{\Omega_0} = \frac{1}{1+\frac{Q_0 t}{J\Omega_0}}$
- Hence, high moment of inertia, low torque, high initial rotor speed help keep angular velocities higher



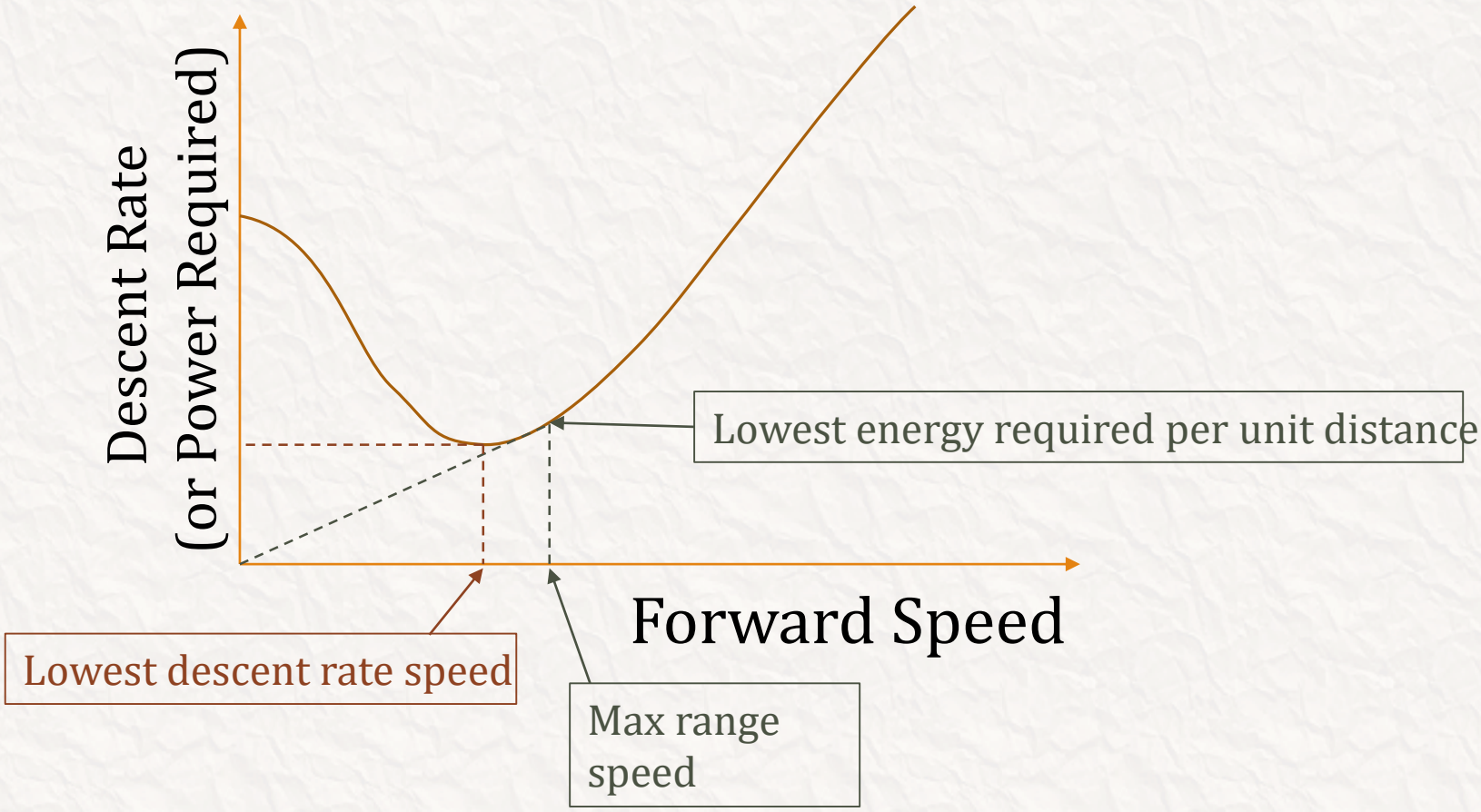
Descent Rate

- Required power supplied by the air (loss in P.E of the vehicle)

  Power Required = Power available = Descent rate × Gross Weight

- $$Descent\ Rate = \frac{Power\ Required}{Gross\ Weight}$$

Descent Rate



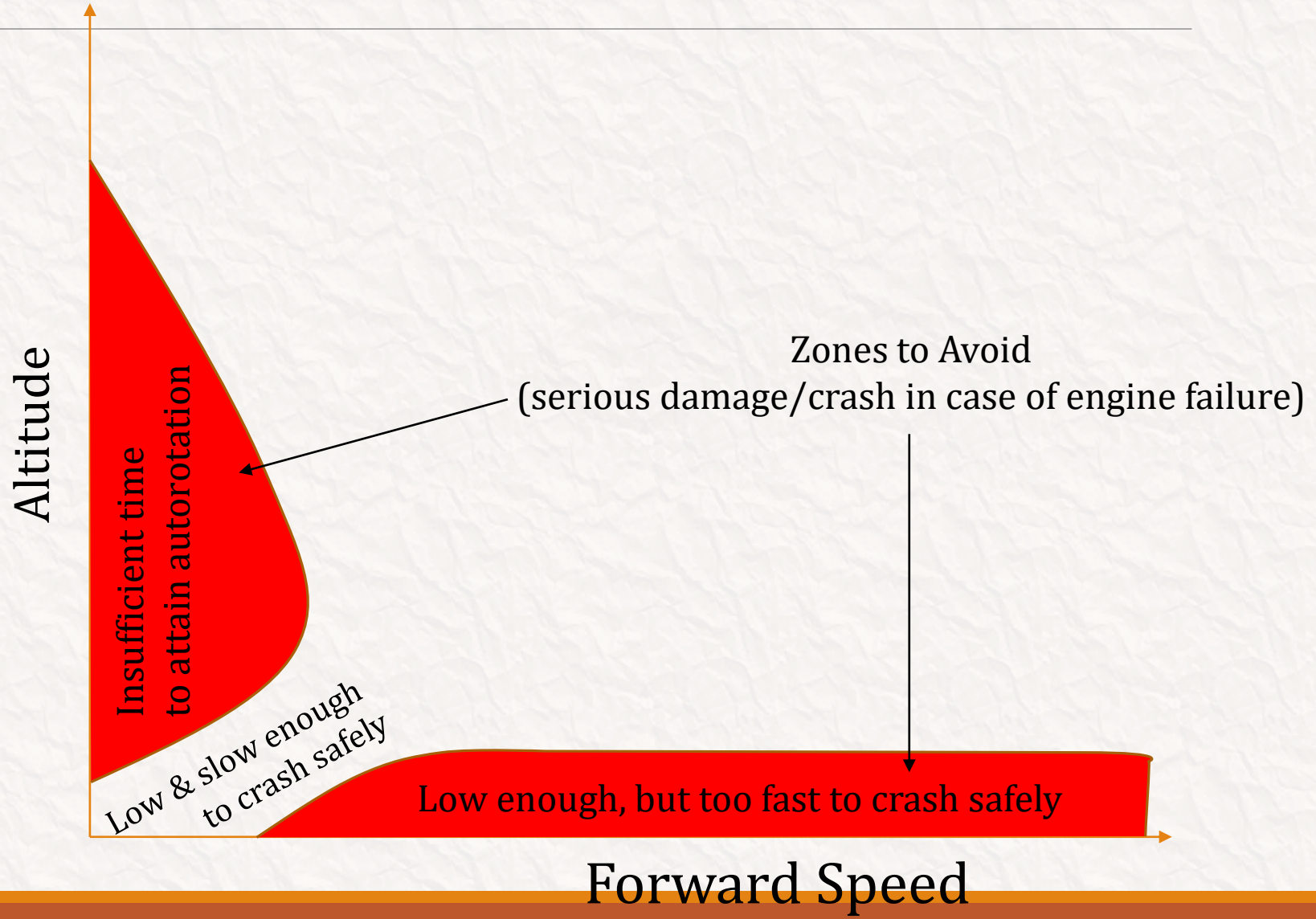
Zoom Manoeuvre

Used when engine failure occurs during high forward speed but low altitude

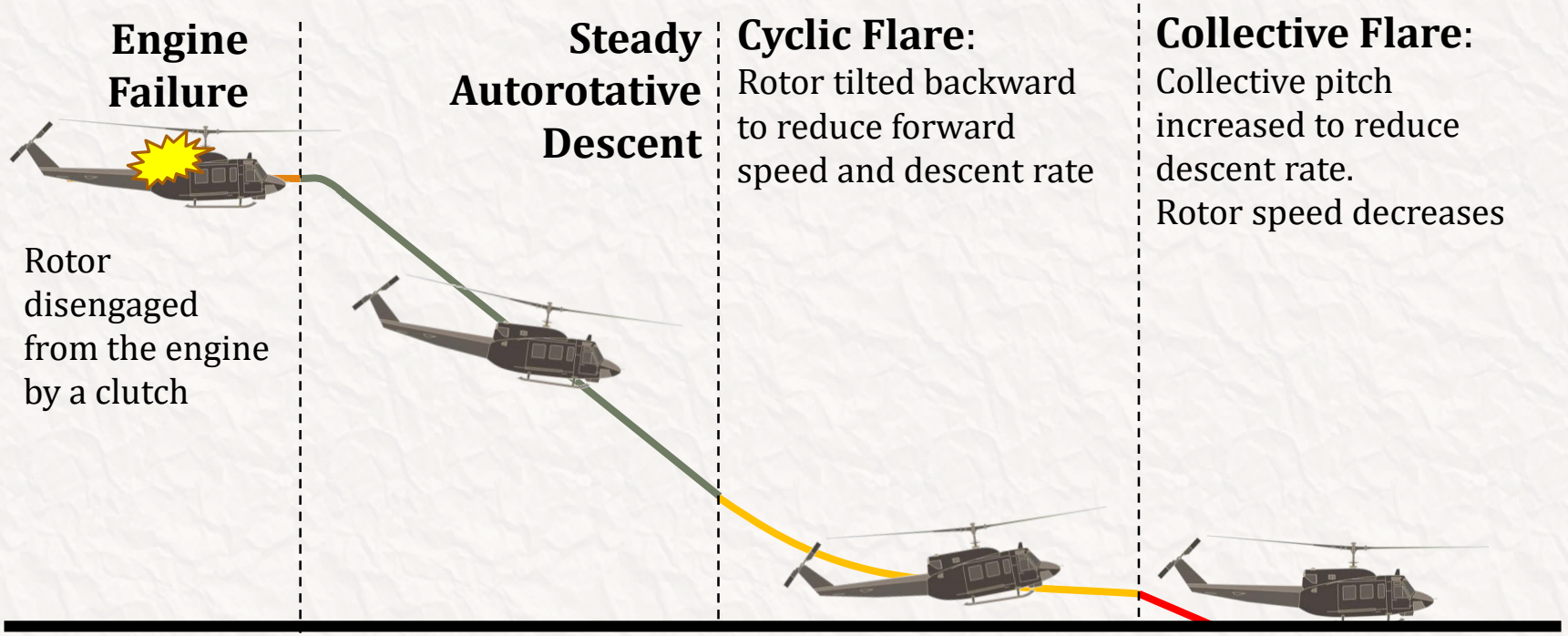
- Pilot trades K.E for P.E to rise to higher altitude
- Followed by the typical autorotative descent.



Dead man's curve



Helicopter Autorotative Landing



Autorotative Index

- A number indicating how well can a helicopter perform autorotative landing flare

- $AI = \frac{J\Omega^2}{GW}\frac{\rho}{\rho_0}\frac{A}{T}$

(need > 60 for single engine helicopters)
(need > 25 for twin engine helicopters)

Max acceleration in level flight

- Find max fuselage equivalent flat plate area (f_{max}) at which steady level flight is still possible (power required = power available)
- Compute max forward directed thrust = $\frac{1}{2}\rho f_{max} V_\infty^2$
- Compute actual thrust needed = $\frac{1}{2}\rho f_{actual} V_\infty^2$
- Excess thrust = max thrust – actual thrust
  - $\frac{1}{2}\rho V_\infty^2 (f_{max} - f_{actual})$
- Max acceleration = Excess thrust/Mass

Max Deceleration

- Rotor disk is turned backward for deceleration
- Strong deceleration can lead to autorotation, which can increase rotor speed, which beyond a limit can damage structure
- Only 10% - 20% rotor over-speeding is permitted
- Compute T, $\alpha_{TPP,}$ H force, Fuselage Drag, etc. at the highest permitted rotor speed in autorotative conditions
- Rearward force = $-T\sin\alpha_{TPP} + H$ + fuselage drag
- Max deceleration = rearward force/mass

